Agentic AI Programme

Curriculum Outline & Speaker Matrix — 2026 Edition

This document provides the full session-by-session curriculum for the **Agentic AI Programme**, including content summaries, learning objectives, and speaker classifications. Each module is mapped to a delivery format and an expert domain. The programme is structured across four thematic pillars: Foundations, Architecture, Regulatory Considerations, and Applied Practice.

1. Programme Overview

The programme spans 12 weeks and covers both theoretical foundations and practical applications of agentic AI systems. Sessions alternate between industry expert guest lectures and hands-on workshops. Assessment is continuous, with a capstone project in Week 11–12.

1.1 Programme Structure

Module	Weeks	Format	Delivery
AI Foundations	1–2	Lecture + Workshop	Hybrid
Agentic AI Architecture	3–5	Lecture + Lab	In-Person
Regulatory & Ethics	6–7	Seminar	Virtual
Frameworks & Tooling	8–9	Workshop	In-Person
Applied Projects	10–12	Project-Based	Hybrid

2. Session-by-Session Curriculum

The table below describes the content of each session, the topics covered, key learning outcomes, and the category of speaker. Speaker categories are: *Academic Expert*, *Industry/Startup Expert*, and *Regulatory/Policy Expert*.

2.1 Module 1: AI Foundations (Weeks 1–2)

Topic	Content	Speaker
Introduction to LLMs and Foundation Models	- History and evolution of language models: from n-grams to transformers. - Attention mechanisms, RLHF, and instruction tuning. **Key architectures:** 1. GPT series, Claude, Gemini, Llama 2/3, Mistral. - Prompt engineering: zero-shot, few-shot, chain-of-thought. - Evaluation benchmarks (MMLU, HellaSwag, HumanEval).	Academic Expert
Agentic AI – From LLMs to Autonomous Agents	- From passive LLMs to agentic behavior: memory, planning, and tool use. - Prompt chaining and self-reflection mechanisms. **Frameworks for agentic AI:** 1. LangChain, AutoGen, CrewAI, OpenDevin, and Semantic Kernel. Multi-agent coordination and emergent behavior. **Agent memory architectures:** 1. Episodic, semantic, and vector-based retrieval. Safety, alignment, and interpretability challenges in agentic AI.	Industry/Startup Expert
Retrieval-Augmented Generation (RAG)	- Dense vs sparse retrieval: BM25, ColBERT, DPR. - Vector databases: Pinecone, Weaviate, Chroma, Qdrant. **Advanced RAG patterns:** 1. HyDE, self-query, multi-hop reasoning. - Chunking strategies and embedding model selection. - Evaluation: RAGAS, TruLens.	Academic Expert

2.2 Module 2: Agentic AI Architecture (Weeks 3–5)

Topic	Content	Speaker
Tool Use and Function Calling	- OpenAI function calling, Anthropic tool use, Gemini tool integration. **Tool categories:** 1. Code execution, web search, database queries, API calls. 2. File I/O, browser automation, calendar and email. - Tool selection and routing strategies. - Error handling and retry logic in agentic loops.	Industry/Startup Expert

Topic	Content	Speaker
Multi-Agent System Design	- Orchestrator-agent and peer-to-peer architectures. **Coordination patterns:** 1. Sequential, parallel, hierarchical. 2. Reflection and critic patterns. - Message passing protocols and shared memory. - Failure modes: hallucination cascades, infinite loops. **Frameworks:** 1. AutoGen Studio, CrewAI, LangGraph.	Industry/Startup Expert
Planning and Reasoning in Agents	- ReAct (Reason + Act) paradigm and chain-of-thought prompting. **Planning algorithms:** 1. Tree-of-Thoughts (ToT), Graph-of-Thoughts (GoT). 2. MCTS-based planning for complex tasks. - Task decomposition and sub-goal generation. - Self-correction loops and reflective agents.	Academic Expert
Observability and Evaluation	- Tracing agent execution with LangSmith, Arize, W&B Weave. **Evaluation metrics:** 1. Task completion rate, step efficiency, tool accuracy. 2. Hallucination rate, context faithfulness. - Human-in-the-loop evaluation pipelines.	Industry/Startup Expert

3. Regulatory and Ethics Module (Weeks 6–7)

This module provides an overview of the global AI regulatory landscape and its implications for organisations deploying agentic systems. Guest speakers include policy advisors, legal experts, and compliance practitioners.

Topic	Content	Speaker
EU AI Act and Global Regulatory Landscape	- EU AI Act: risk categories, prohibited practices, and high-risk systems. - GDPR interaction with generative AI systems. **Comparative regulatory review:** 1. US Executive Order on AI Safety. 2. UK Pro-Innovation Regulatory Approach. 3. India DPDPA 2023 and proposed AI Act. - Conformity assessments and CE marking for AI.	Regulatory/Policy Expert
AI Safety, Alignment, and Responsible Deployment	- Constitutional AI and RLHF-based alignment techniques. **Safety properties for agentic systems:** 1. Corrigibility, value alignment, and human oversight. 2. Minimal footprint and reversibility principles. - Red-teaming, adversarial testing, and jailbreak resistance. - Bias, fairness, and model transparency auditing.	Academic Expert
Data Privacy and Intellectual Property	- Training data provenance and copyright implications. **Data minimisation principles for AI:** 1. Purpose limitation and storage constraints. 2. Subject access rights in AI-generated outputs. - Watermarking, provenance tracking, and content authentication. - Liability frameworks for autonomous AI decisions.	Regulatory/Policy Expert

4. Assessment and Evaluation Framework

Participants are evaluated on three dimensions: conceptual understanding, practical implementation, and critical reflection on regulatory implications.

Component	Weight	Week Due	Description
Quiz — Foundations	10%	Week 2	Multiple choice, open-book, 45 minutes.
Architecture Design Report	20%	Week 5	Design a multi-agent system for a given use-case (1500 words).
Regulatory Impact Assessment	20%	Week 7	Analyse EU AI Act applicability to a provided system specification.
Framework Implementation	20%	Week 9	Build a working agentic pipeline using LangGraph or AutoGen.

Capstone Project	30%	Week 12	End-to-end agentic system with documentation, evaluation, and demo.

[1] All sessions are recorded. Participants must comply with the recording consent notice issued at programme start. Recordings are stored for 12 months post-programme and accessible only to enrolled participants.
[2] Speaker engagements are subject to availability. Birchlogic AI Labs reserves the right to substitute with an equivalent expert without prior notice.
[3] Content in this curriculum is updated annually. Refer to Table 1 (Programme Structure) for the official module sequence. See Table 3 (Regulatory Module) for data privacy obligations.